



ICT 1209 - Web Technology

Final Mini Project Report

Smart Student Resources Hub

Student Resources Hub

Group No – 12

Group Members

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Date – 2024/9/27	

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Project Overview

Purpose

The primary purpose of the "Smart Student Resource Hub" is to create a centralized, user-friendly digital space where students can access, organize, and share educational materials such as lecture notes, past papers, and academic announcements. Currently, students waste valuable time searching for files scattered across multiple platforms like WhatsApp groups, emails, and physical copies. The goal of this system is to eliminate that fragmentation by bringing everything into one unified dashboard. Ultimately, this reduces academic stress, improves learning efficiency, and fasters a collaborative peer-to-peer learning environment.

Target Audience

The "Smart Student Resource Hub" is designed to serve a diverse demographic within the academic ecosystem. While primarily tailored for higher education, its versatile architecture allows it to benefit various stakeholders who require a structured and collaborative environment for academic success. The platform caters to the following key groups

1. University Students

As the core demographic of the platform, university students require a highly organized digital space to navigate their complex academic schedules. The hub provides them with a structured environment to easily locate semester-wise study materials, upload their own custom study guides, and systematically track their academic resources. By centralizing these files, students can focus more on learning rather than searching for lost documents across multiple messaging groups.

2. Lecturers and Academic Staff

Beyond students, the platform acts as a vital administrative and teaching tool for educators. Lecturers can utilize the hub to seamlessly distribute official course materials, broadcast important academic announcements, and ensure that all enrolled students have equitable access to the correct, up-to-date resources. This streamlines communication and reduces the friction of sharing large files via email.

3. A/L Students

The platform also extends its educational benefits to high school students preparing for their critical Advanced Level examinations. By accessing a centralized repository of reference materials, past papers and comprehensive study guides shared by senior educational communities A/L students can gain valuable insights and high-quality resources to better prepare for university level education.

4. Educational Communities and Study Groups

Finally, the hub serves as a digital gathering space for academic clubs, study groups, and wider educational communities. It is built to facilitate seamless knowledge sharing, encourage peer-to-peer discussions, and promote collaborative learning initiatives that extend far beyond the boundaries of a traditional physical classroom.

Key Features

1. Secure User Login and Registration System

Provides personalized accounts with secure authentication. Users have custom profiles to manage their details and personalized dashboards.

2. Resource Upload and Download Management

A structured system allowing users to upload study materials in various formats (PDF, DOCX, ZIP, Excel). Includes a dynamic storage tracker to monitor usage and a “History” tab to keep track of previously accessed or uploaded documents.

3. Advanced Search and Filtering Engine

A robust search mechanism that allows users to find specific materials quickly. Users can filter resources by categories Ex : Technology, Education, Environment file type, or date, making navigation highly efficient.

4. Announcement and Notification Section

An integrated messaging and notification system to keep users updated on newly added resources, important academic deadlines, and system announcements.

5. Personalized User Dashboard

A centralized hub displaying real-time statistics like storage usage, saved/favorite documents, recent history, and quick access links to enhance the overall user experience (UX).

System design

Database schema

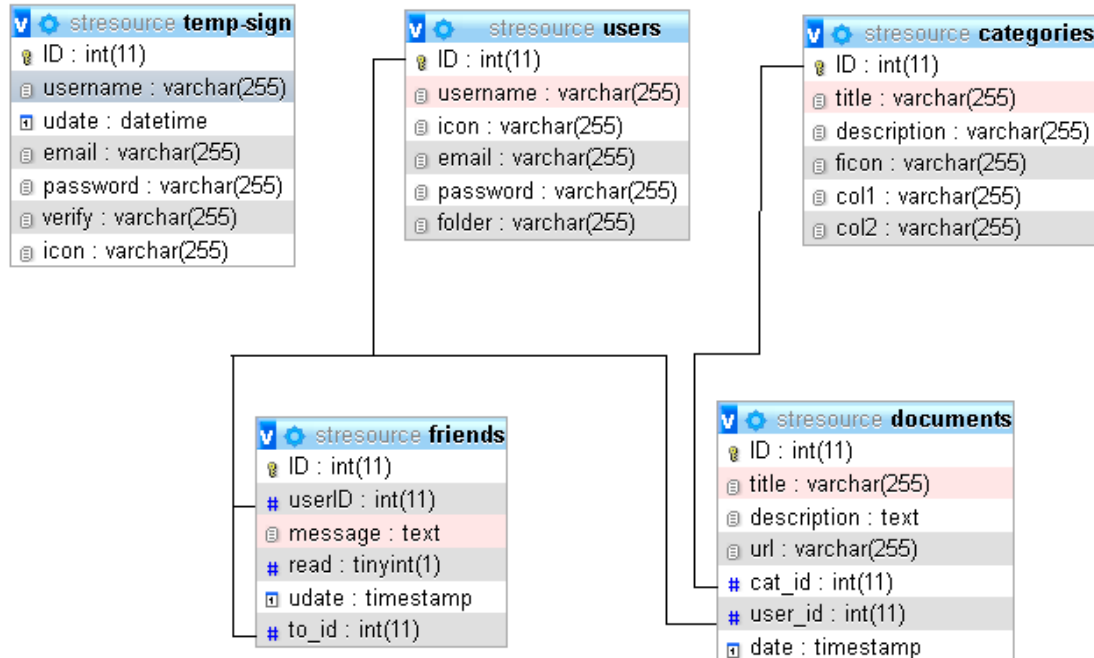


Figure 1: Database schema

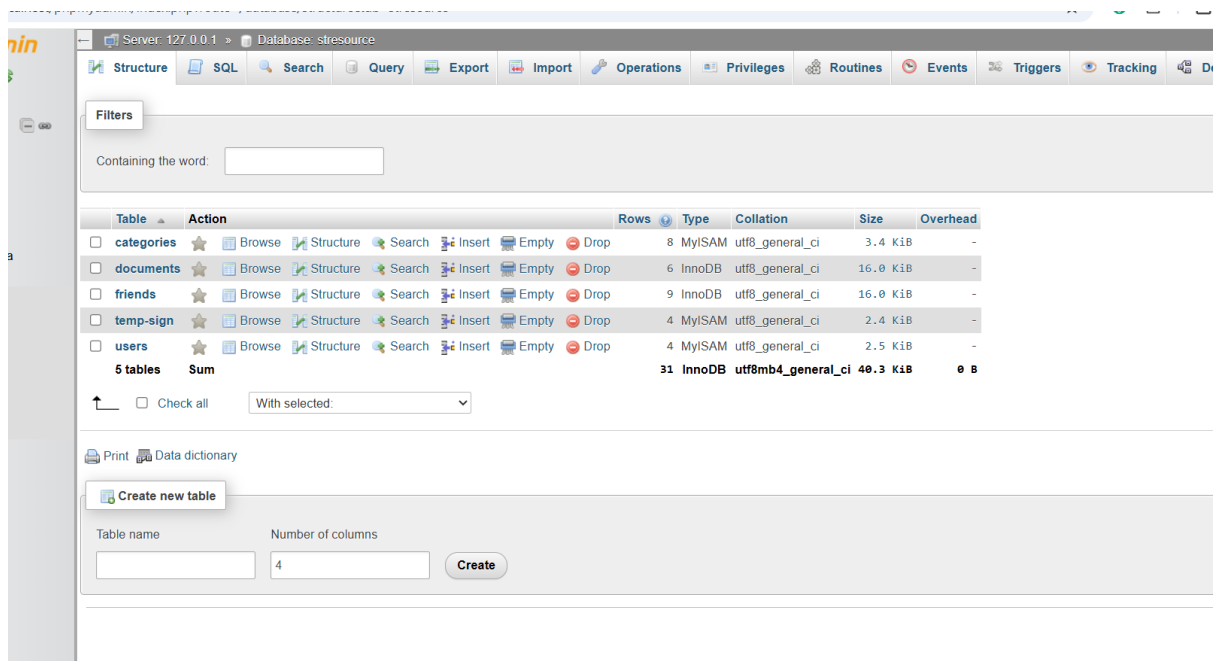


Figure 2: Database

Folder structure

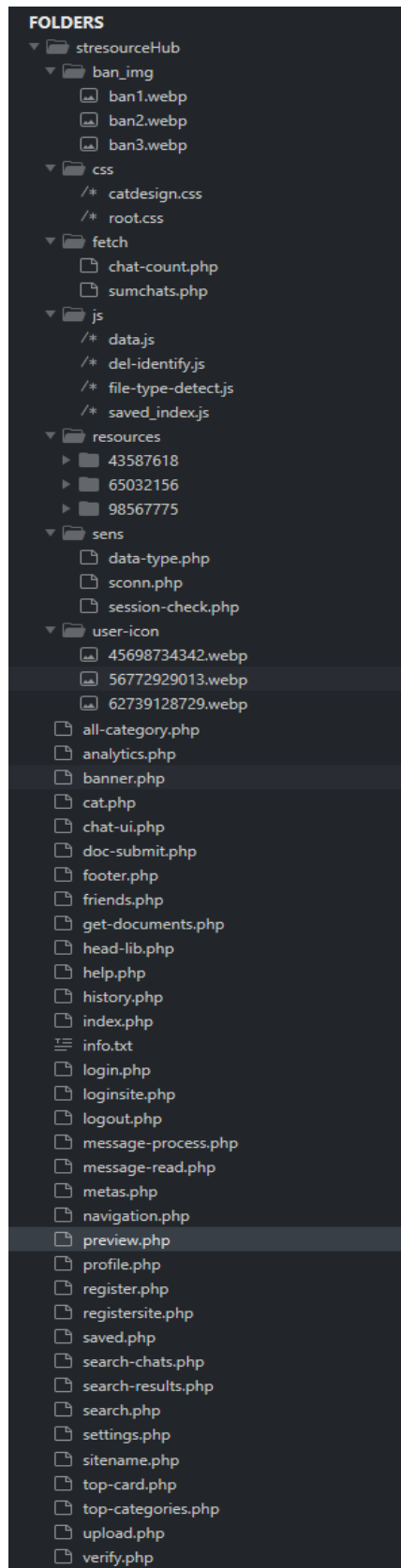


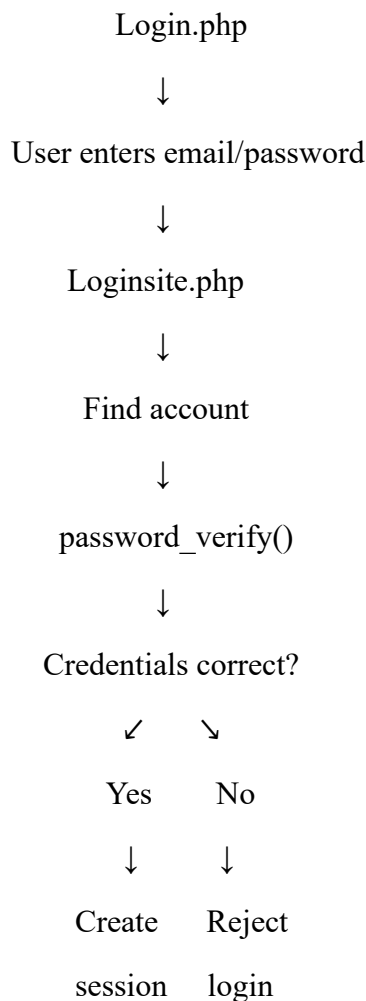
Figure 3: Folder structure

Implementation

This section outlines the detailed implementation of the major features within the "Smart Student Resource Hub". The front-end architecture was meticulously designed using HTML, CSS, Bootstrap 5 and JavaScript to ensure a highly responsive and fluid user experience. The design process heavily prioritized User-Centered Design (UCD) principles and usability heuristics to ensure the platform is intuitive for diverse user groups ranging from university students to academic staff.

1. Login Interface

The authentication portal acts as the secure gateway to the platform. To minimize cognitive load and adhere to minimalist design principles the interface uses a clean, card-based layout centered on the screen. It incorporates client-side form validation to prevent input errors before they reach the server providing immediate visual feedback to the user. Furthermore to ensure a seamless user journey for unauthenticated users, a prominent "New Registration" routing button is strategically placed directly below the primary login action. The interface also fully integrates the global navigation header and the comprehensive site footer ensuring users maintain contextual awareness and have immediate access to support contacts and quick links even prior to authenticating.



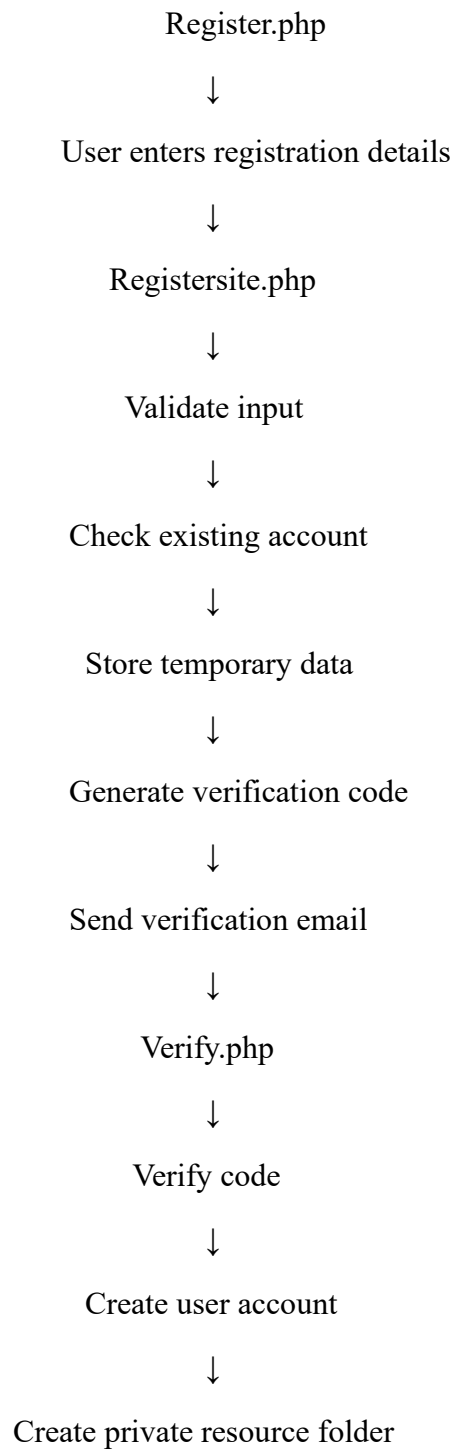
↓
Protected pages
↓
session_check.php

The screenshot displays the login interface of a web application named 'ResourceHub'. The header is a gradient of pink and orange, containing a menu icon, the site name, and a user profile. The login form is a white card with a title, a user icon, and two input fields for email and password. A green 'Login' button and a 'New Registration' link are also present. The footer is dark grey and contains four columns of information: 'About Us', 'Quick Links', 'Contact', and 'Follow Us'.

Figure 4: logging

2. User Registration

The "Create Account" interface serves as the primary onboarding point for new students and educators. To maintain design consistency and reduce cognitive friction it utilizes a clean centered card-based UI that mirrors the login portal. The registration form captures essential user data (Username, Email address and Password) and is engineered to incorporate client-side validation ensuring data integrity before server submission. To further streamline the onboarding process and reduce form fatigue a "Register with Google" Single Sign-On option is strategically provided. Additionally this page fully integrates the global navigation header and the comprehensive site footer ensuring users maintain contextual awareness and have immediate access to support contacts and quick links during the registration phase.





ResourceHub

 Hello123

Create Account



Username

Enter username

Email address

Enter email

Password

Enter password

Register

or

 Register with Google

About Us

We provide a collaborative hub for sharing resources, knowledge access and community connection

Quick Links

- > Home
- > Login
- > Register
- > Categories

Contact

Colombo, Sri Lanka

0702473866 | 0721020034

chamudupravishkka@gmail.com
dushaneranda0803@gmail.com

Follow Us

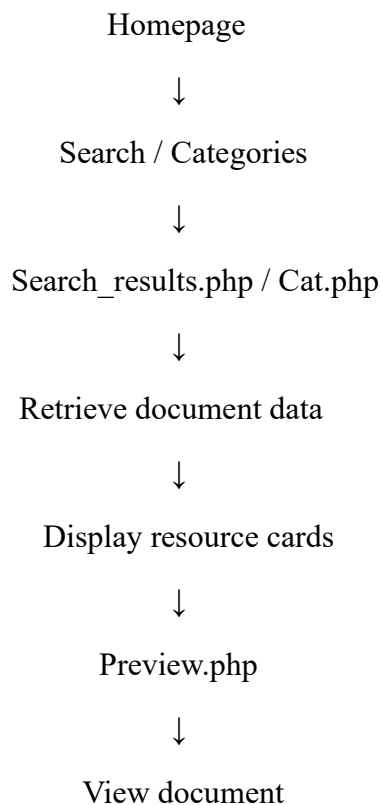


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Figure 5:Registre

3. Interactive Home Page & Resource Discovery Engine

Designed as the primary navigation hub the Home Page focuses on "Recognition rather than Recall" by using highly visual elements. It features a dynamic responsive hero carousel that can be utilized to broadcast critical university announcements or featured educational content. Below the fold a robust dual-search mechanism is implemented. It combines a universal text search bar with advanced filtering alongside categorical quick access cards (PDF, Word, Excel, PPT, ZIP). By grouping resources into broader domain categories such as Technology, Environment and Education accompanied by distinct vector graphics the system guides users efficiently drastically reducing the time spent searching for specific past papers or lecture notes.



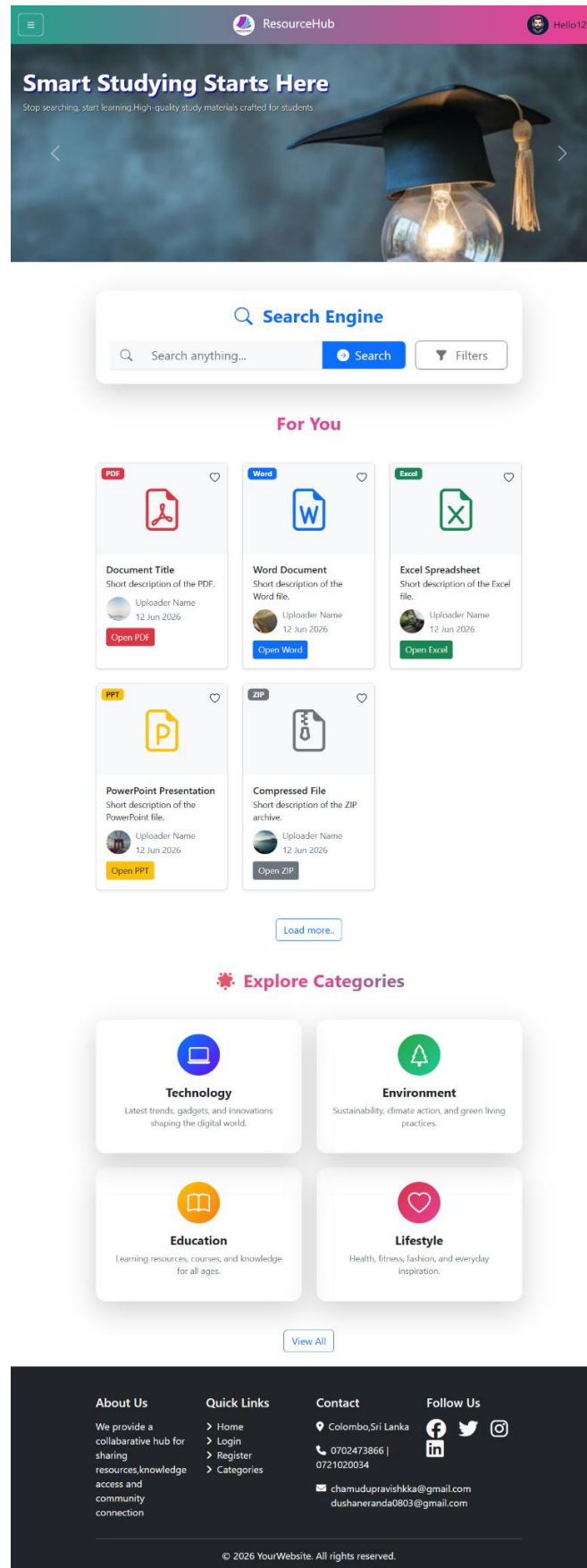


Figure 6: Home page

4. Personalized User Dashboard

The user dashboard operates as a personalized command center, built upon a responsive grid system that adapts perfectly to both mobile and desktop environments. A critical usability feature here is the "Visibility of System Status." This is achieved through a real-time "Storage Usage" progress bar, which provides immediate, visually digestible feedback regarding the user's data consumption limits. The dashboard utilizes a modular card design to compartmentalize information, featuring alert badges for new messages and strategically placed quick action buttons (Settings, Help, Analytics, Logout). This flat navigation structure minimizes click-depth and enhances overall user efficiency.

My Dashboard



Choose a Feature



↓	↓	↓	↓
Settings	Help	Analytics	Logout
↓	↓	↓	↓
Manage Account	Get Help	View Usage Statistics	Exit System

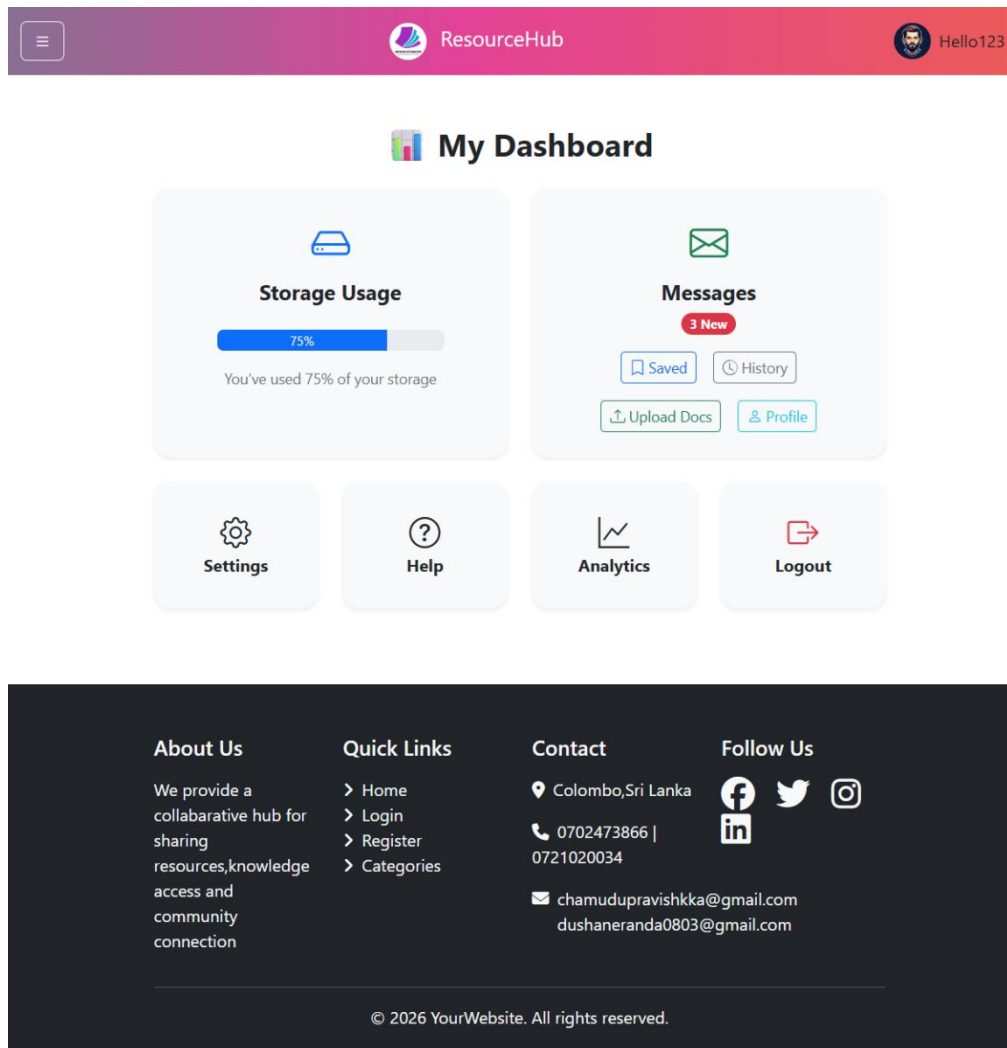
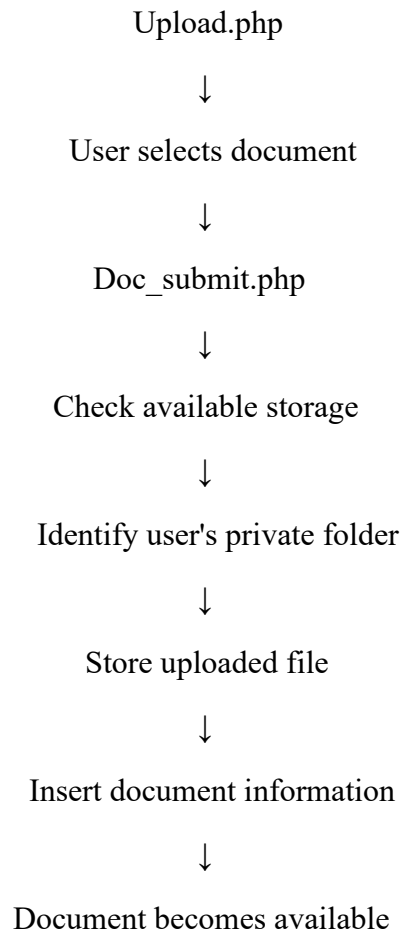


Figure 7: Dashboard

5. Advanced Document Upload Management

The document ingestion interface is engineered to facilitate seamless file sharing between peers and lecturers. It integrates a robust file picker that implicitly enforces file-type validation restricting uploads to supported formats like PDF, DOCX, JPG, PNG thereby maintaining database integrity and preventing malicious uploads. Providing users with maximum control and freedom, the interface features a dynamic category management system. Users are not restricted to pre-existing folders; they can seamlessly declare and append a "New Category" directly from the upload panel using asynchronous behavior, eliminating the need for disruptive page reloads.



 ResourceHub Hello123

 **Upload Document**

Document Title

Description

Choose File

Supported formats: PDF, DOCX, JPG, PNG

Select Category

Create New Category

About Us

We provide a collaborative hub for sharing resources, knowledge access and community connection

Quick Links

- > Home
- > Login
- > Register
- > Categories

Contact

Colombo, Sri Lanka

0702473866 | 0721020034

chamudupravishkka@gmail.com
dushaneranda0803@gmail.com

Follow Us

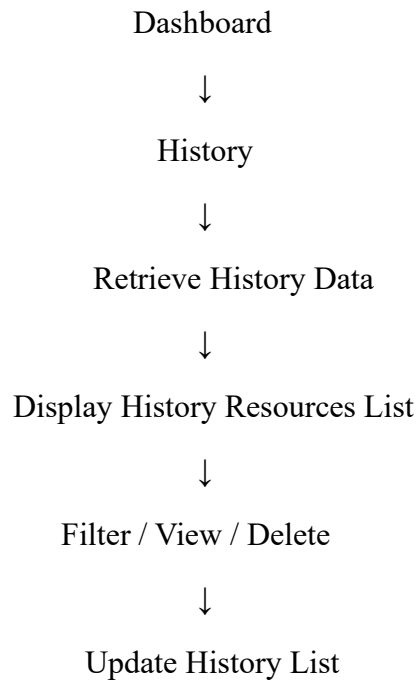
Figure 8: upload

6. History and Saved Items Modules

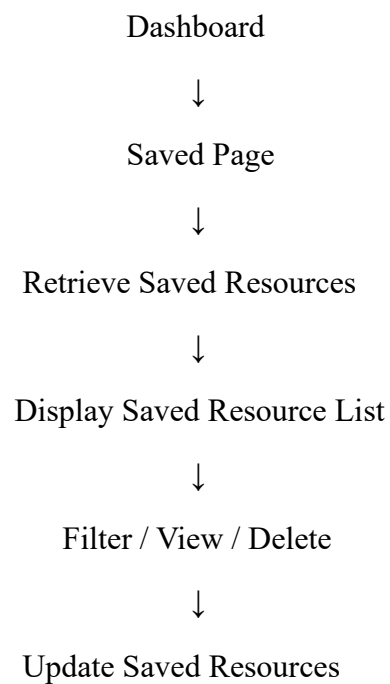
To give users comprehensive control over their digital footprint and study materials, the platform incorporates dedicated "History" and "Saved" modules. Both modules present complex datasets using clean, responsive data tables. To manage extensive lists of academic files, a built-in calendar date-picker tool allows for precise chronological filtering. Each record clearly displays the document name, a color-coded file type badge and a timestamp. Furthermore, individual "Delete" action buttons are integrated into

each row, allowing users to effortlessly curate their saved lists and maintain data privacy.

History Interface



Save Interface



ResourceHub

Hello123

History

Filter by Date

mm/dd/yyyy

Apply

☐	History Link	Type	Time	
☐	Document 1	PDF	01 Jul 2026	
☐	Image 1	Image	29 Jun 2026	
☐	Report	Doc	12 Jun 2026	

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Contact

Colombo, Sri Lanka

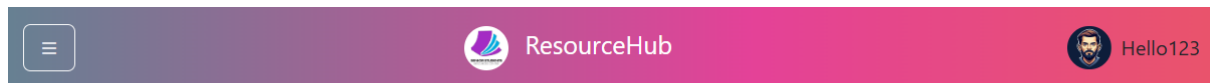
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Figure 9: History



Saved

Filter by Date

mm/dd/yyyy



Apply

☐	Saved Link	Type	Time	
☐	Document 1	PDF	01 Jul 2026	
☐	Image 1	Image	29 Jun 2026	
☐	Report	Doc	12 Jun 2026	

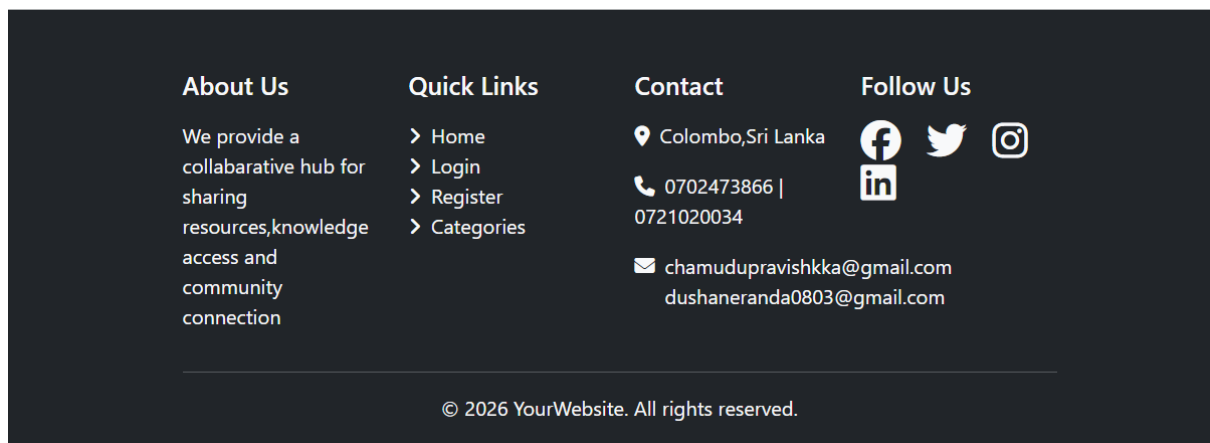


Figure 10: Save

7. System Settings and Real-Time Analytics Visualizations

The Settings panel goes beyond standard profile management by offering granular administrative controls and unique system insights. It provides real-time transparency by displaying the user's current online status, browser engine details and estimated network speeds, alongside hardware metrics like RAM and CPU allocation. To further support data driven user habits, the "Analysis Graph" module employs dynamic data visualization techniques. By translating raw backend storage and history data into interactive line charts the system significantly reduces the user's effort to comprehend their platform activity trends over time.

Dashboard



Setting Page



Load User Settings



Display Account Details



Display System & Network Information



Display Device Storage Information



Quick Links



Update Account Details / Manage History / Manage Saved Items / Help & Support



Update Settings

ResourceHub

Hello123

Settings

← Back to Dashboard

Account Details

FULL NAME

EMAIL ADDRESS

Update Details

System & Network

Online Status

● Online

Browser Info

Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/152.0.0.0 Safari/537.36 Edg/152.0.0.0

Network Speed

1.4 Mbps (3g)

Device Storage

Storage Used	0.00 GB
Storage Available	10.00 GB
Storage Usage	0.0%
RAM	8 GB
CPU Cores	12 Cores
Operating System	Windows
Platform	Win32
Screen Resolution	1536 x 864
Language	en-GB
Time Zone	Asia/Colombo

Quick Links

Manage History

>

Manage Saved Items

>

Help & Support

>

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- > Register
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 dushaneranda0803@gmail.com

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Figure 11: Setting

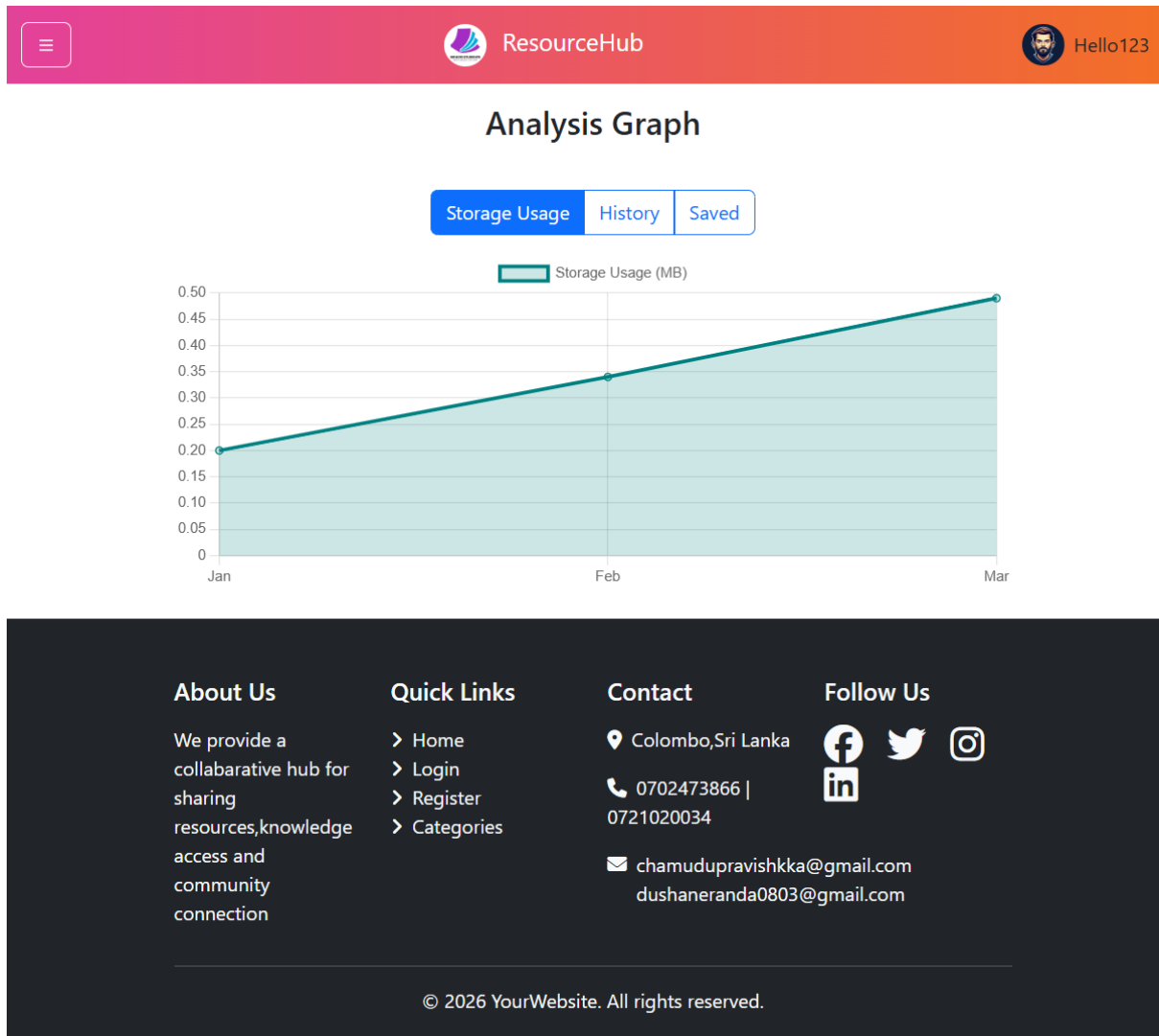
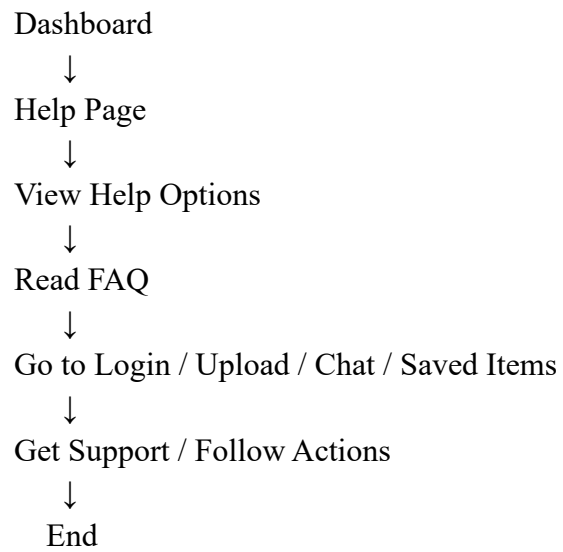


Figure 12: Analysis

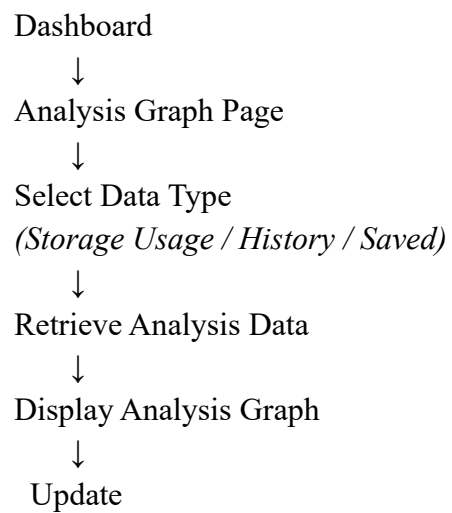
8. Help Interface

Recognizing that systems should be easy to use without documentation, but providing it is crucial when necessary, the "Help" interface strictly adheres to Nielsen's "Help and Documentation" usability heuristic. It features a visually distinct, card-based layout offering quick, step-by-step tutorials for core functionalities Ex: How to Login, Upload, Chat. Each guide includes a direct Call to action button to immediately navigate the user to that specific feature. Furthermore a Frequently Asked Questions section is implemented using interactive collapsible accordion menus. This "progressive disclosure" technique prevents information overload keeping the UI clean while ensuring comprehensive support is readily accessible.

Help Interface



Analysis Interface



 ResourceHub Hello123

? Help

Learn how to use the site step-by-step.

[← Back to Dashboard](#)



1. How to Login

Enter your email and password on the login page. You can also sign in with Google for quick access.

[Go to Login →](#)



2. How to Upload

Click on 'Upload Docs' button in Messages/Dashboard. Choose your PDF or file and click upload to store safely.

[Upload Now →](#)



3. How to Chat

Go to Messages section to view new notifications, read unread messages, and send quick replies.

[Open Messages →](#)



4. How to Save Items

Click the heart or bookmark icon on any document card to save it for later. Access all from 'Saved Items'.

[View Saved →](#)

i Frequently Asked Questions (FAQ)

How do I check or update my account details ▲

You can easily check or update your account from the Settings page.
[Go to Settings Page →](#)

What file formats are supported for upload? ▼

How do I update my remaining storage usage? ▼

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Figure 13: Help

9. Chat Directory Interface

To manage peer-to-peer interactions, the platform features a comprehensive chat directory. To enhance the "Flexibility and Efficiency of Use," this interface integrates a dynamic search bar and a dropdown filtering system, enabling users to rapidly locate specific conversations or study partners without scrolling through long lists. The conversation list employs a strong visual hierarchy displaying rounded user avatars, name identifiers, message preview snippets and precise timestamps. Crucially attention-driven design is utilized via vibrant notification badges to immediately alert users to pending academic discussions.

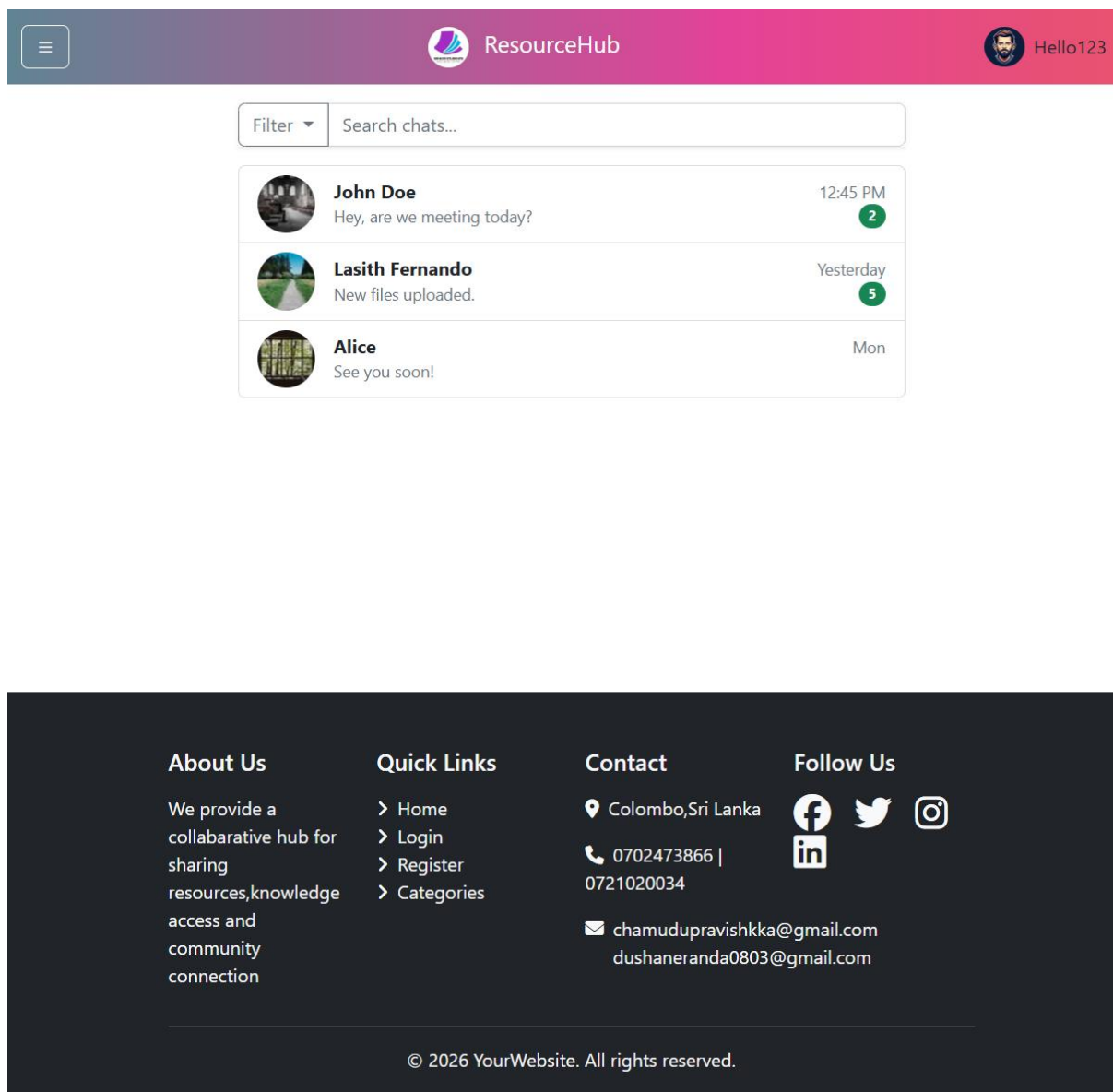
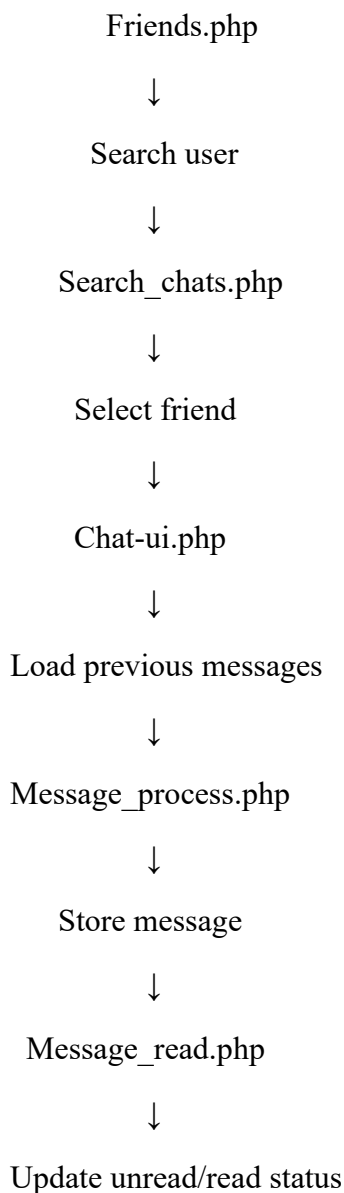


Figure 14: Friend

10. Chat Interface

Fostering a collaborative learning environment is a core objective of the platform, achieved through this dedicated direct messaging interface. The UI deliberately adopts universally recognized chat design patterns utilizing distinct, color-coded and aligned message bubbles for sent versus received texts. This matching between the system and the real world minimizes the user's learning curve. A real-time "Live" status badge provides clear visibility of the system's status while a streamlined text input area with a prominent send icon ensures frictionless communication between students study groups, and lecturers.



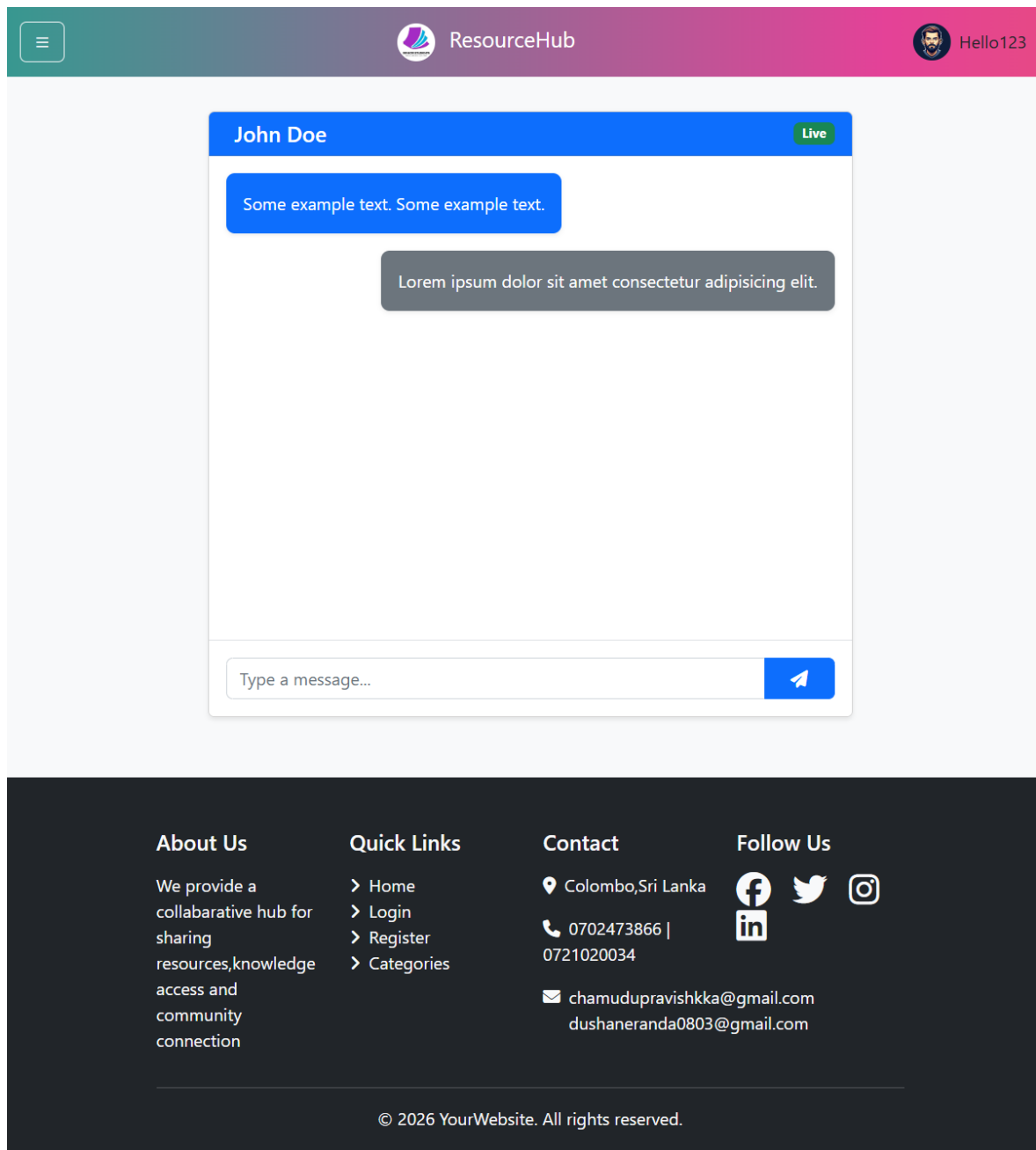


Figure 15: chat

Challenges and solutions

The development of the Smart Student Resource Hub used different frontend and backend technologies. Building a platform with many useful features that is also easy for everyone to use was challenging. Our team faced several technical and design problems during development. We found suitable solutions to make the application secure, responsive, and user-friendly.

Challenge 01 : Writing traditional database connection and query processing code repeatedly in many PHP files would make the application larger and harder to maintain.

Solution : Reusable functions were developed in "Sens/sconn.php", including:

"getData()"

"setData()"

"getDataWithJoins()"

This provides a consistent method for database operations across the application.

Challenge 02 : User-uploaded resources should not simply expose their physical storage location.

Solution : Each user receives an individual resource folder and the application generates the required resource access path when documents are requested.

Challenge 03 : Different pages require different levels of access. Some pages can be publicly accessed while others require a logged in user.

Solution : The "session_check.php" system uses session information together with the "\$allow" variable to control page access.

Challenge 04 : User passwords must not be stored as plain text.

Solution : The registration and login system uses password hashing and "password_verify()" for password verification.

Challenge 05 : A registration process needs to verify the user's email before creating the permanent account.

Solution : Temporary registration data is stored in the "tempsign" table, and a verification code/link is sent to the user's email. "Verify.php" validates the code before creating the final account.

Challenge 06 : Users need to see updated unread message counts and chat information without manually refreshing the page.

Solution : JavaScript polling is used to periodically request updated information from files such as:

"Fetch/chat_count.php"

"Fetch/Sumchats.php"

"Message_read.php"

Challenge 07 : Users should not be able to upload unlimited amounts of data.

Solution : The system provides a 1 GB storage limit per user and checks the user's available storage before processing a new upload.

Individual contribution

The successful completion of the "Smart Student Resource Hub" was a collaborative effort, with responsibilities strategically divided to ensure efficient development. Both members contributed equally to the core backend logic and the comprehensive documentation of the project.

1. D.M.D.E. Bandara (Index: 2807)

- **UI/UX Design** - Helped create wireframes and design user-friendly interfaces using Figma.
- **Frontend Development** - Developed the Login, User Profile, Settings, Help, File Upload, History, Saved Items and Analytics pages.
- **Backend & Documentation** - Helped with the backend structure and prepared the Mini Project Proposal and Final Project Report.

2. B.C.P. Coorey (Index: 2808)

- **UI/UX Design** - Helped design page layouts and improve the user experience using Figma.

- **Frontend Development** - Developed the Home Page, Categories, Document Preview, Search, Registration, Friends/Community and Chat features.
- **Backend & Documentation** - Helped with the backend structure and prepared the Mini Project Proposal and Final Project Report.

Setup Instructions

1. Download and install XAMPP.
2. Open the XAMPP Control Panel and start Apache and MySQL.
3. Copy the project folder into the C:\xampp\htdocs directory.
4. Open a web browser and go to <http://localhost/phpmyadmin>.
5. Create a new database named 'stresource'.
6. Click on "Import" select the database.sql file from the backened/ and click "Go".
7. After import delete the database.sql from the folder.
8. Open a new tab and navigate to <http://localhost/stresource> to run the application.

References

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